Predicting Hospital Readmissions for Diabetes Patients

An End-to-End Predictive Analytics & Machine Learning Project

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Core Objectives & Deliverables

- **Data Acquisition & Governance:** Ingest and audit 100k+ clinical records from the UCI Diabetes repository.
- **Messy Data Wrangling:** Handle missing indicators, treat high-cardinality diagnosis codes, and normalize features.
- **Predictive Modeling:** Train and tune classification pipelines to predict 30-day patient readmissions.
- **Risk Stratification & Insights:** Construct 12 diagnostic visualization dashboards to guide intervention protocols.
- **Full Repository Setup:** Modular codebase structured with automated error logging and an executive summary.

Segment 1: Project Overview & Data Acquisition

Pipeline Stages:

- Business problem formulation
- Source ingestion (UCI Repository)
- Virtual environment configuration
- Data loading and schema validation

Project Profile

Domain: Healthcare Operational Analytics
Problem: Forecast 30-day diabetic readmissions to cut costs and enhance patient care.
Tech Stack: Python, Pandas, Scikit-learn, SQL
Dataset: 130-US Hospitals (100k+ records)

Segments 2 & 3: Preprocessing & Exploratory Analysis

Data Cleaning & Transformation:

- Quality assessment and duplicate resolution
- Imputation of systemic missing values
- Categorical encoding for ICD-9 diagnosis codes
- Outlier evaluation across admission durations

Exploratory Data Analysis (EDA):

- Target class distribution and imbalance analysis
- Bivariate correlations with readmission rates
- Age-bracket and medication impact profiling
- Automated visualization plot generation

Segment 4: Model Building & Evaluation

- **Model Experimentation:** Evaluated Logistic Regression, Random Forest, and Gradient Boosting benchmarks.
- **Performance Evaluation:** Validated through ROC-AUC, precision-recall curves, and confusion matrices.
- **Feature Importance:** Extracted clinical drivers (number of lab procedures, prior inpatient encounters, medication changes).
- **Probability Thresholds:** Adjusted classification cutoffs to minimize false negatives in high-risk patient segments.

Segment 5: Repository Architecture & Artifacts

```
healthcare-readmission-prediction/
   diabetic_data.csv                # Raw dataset
   diabetic_data_cleaned.csv        # Cleaned master data
   healthcare_analytics.py          # Pipeline script
   EXECUTIVE_SUMMARY.txt            # Executive brief
   README.md                        # Project documentation
   requirements.txt                 # Python dependencies
   visualizations/                  # Analytical outputs
       01_target_distribution.png
       02_numeric_distributions.png
       03_categorical_distributions.png
       04_numeric_vs_readmission.png
       05_categorical_vs_readmission.png
       06_correlation_matrix.png
       07_model_comparison.png
       08_confusion_matrices.png
       09_feature_importance.png
       10_probability_distribution.png
       11_risk_stratification.png
       12_FINAL_DASHBOARD.png
```

Deliverables:

- Complete reproducibility with `requirements.txt`
- Modular Python workflow script
- 12 executive dashboard artifacts
- Full GitHub documentation